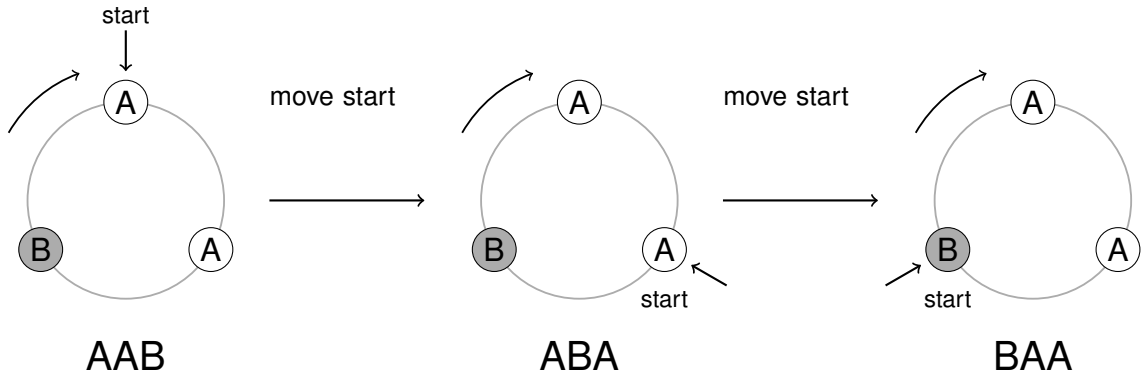
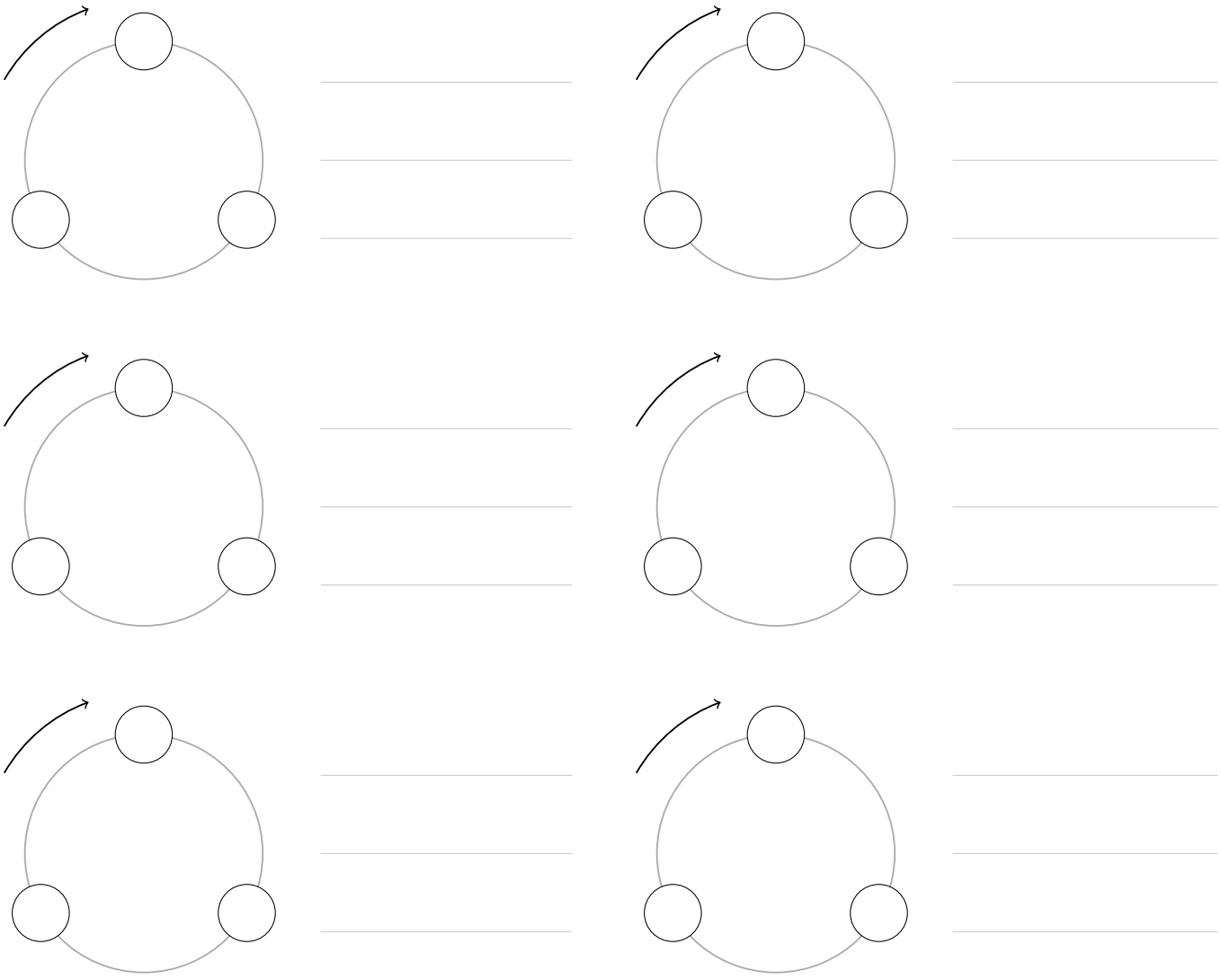
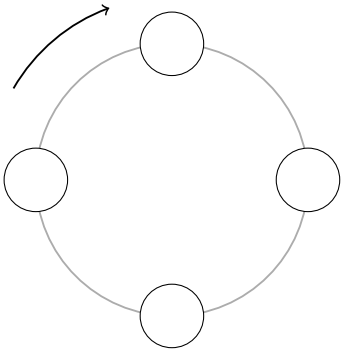
Each bead may be A or B; one-color rings are allowed. Read clockwise from the start arrow. The bead order is the readout. Rings count as the same when turning makes them match. Keep them face up; do not flip them over.

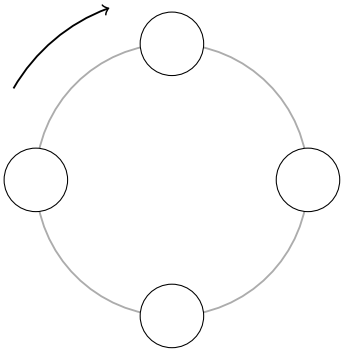


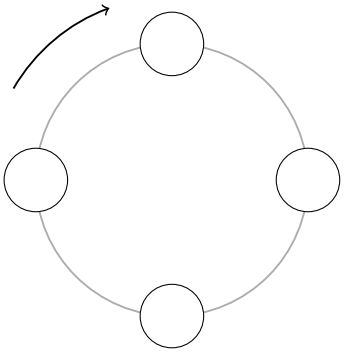
Problem 1: Find every three-bead ring using A and B. How many different readouts does each ring have? Someone divides 8 by 3 to count the rings. What went wrong?

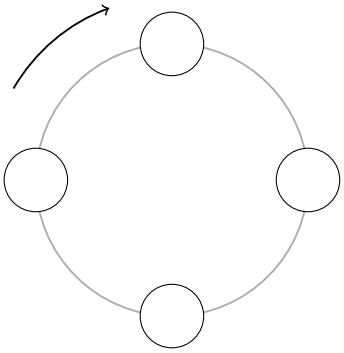


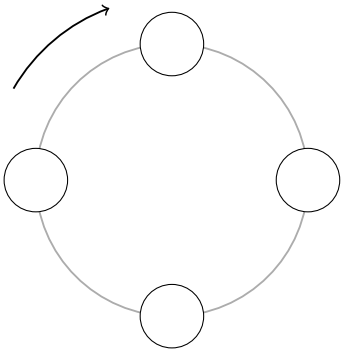
Problem 2: Find four-bead rings with one, two, three or four different readouts, or explain why a requested number is impossible. What feature lets a mixed ring repeat early?

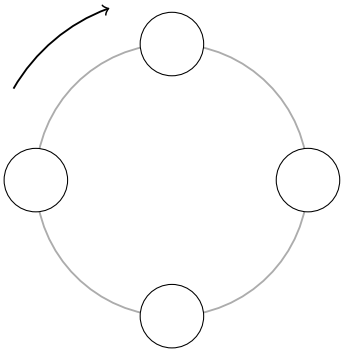






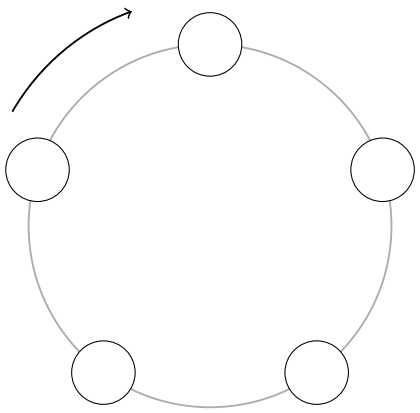




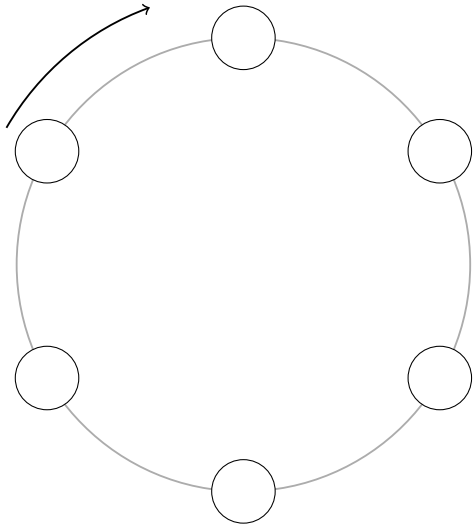


Problem 3: Group the 32 cards by turning. Count the rings with a calculation, and explain why the group sizes differ from the four-bead case.

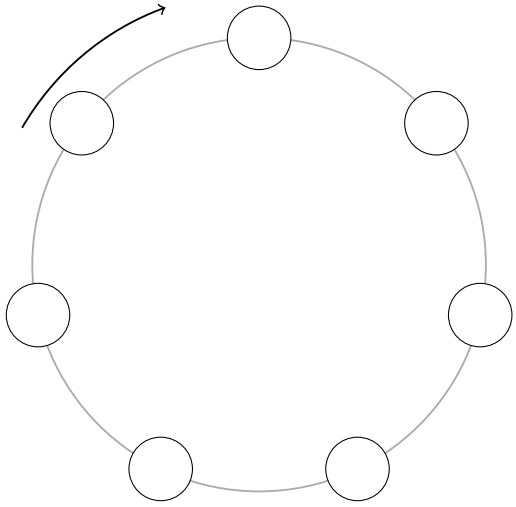
start 1 A A A A A	start 2 A A A A B	start 3 A A A B A	start 4 A A A B B
start 5 A A B A A	start 6 A A B A B	start 7 A A B B A	start 8 A A B B B
start 9 A B A A A	start 10 A B A A B	start 11 A B A B A	start 12 A B A B B
start 13 A B B A A	start 14 A B B A B	start 15 A B B B A	start 16 A B B B B
start 17 B A A A A	start 18 B A A A B	start 19 B A A B A	start 20 B A A B B
start 21 B A B A A	start 22 B A B A B	start 23 B A B B A	start 24 B A B B B
start 25 B B A A A	start 26 B B A A B	start 27 B B A B A	start 28 B B A B B
start 29 B B B A A	start 30 B B B A B	start 31 B B B B A	start 32 B B B B B



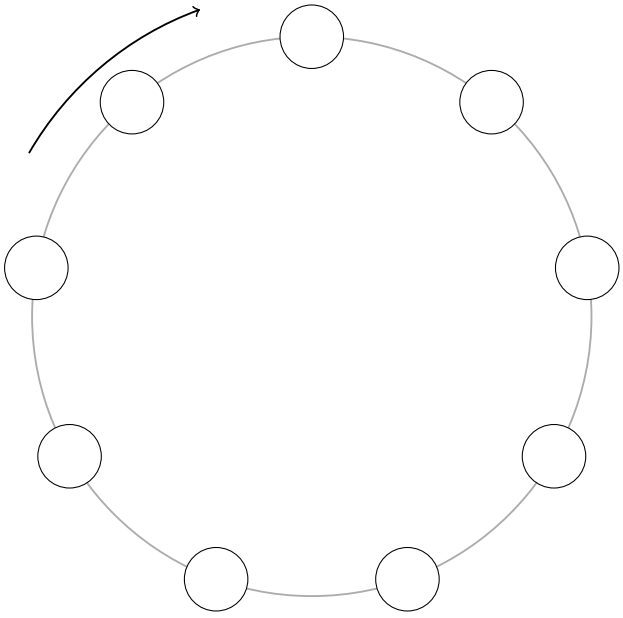
Problem 4: Which numbers of different readouts can a ring using both A and B have on six beads? On seven? On nine?



6 beads



7 beads



9 beads

Problem 5: A prime number has exactly two positive divisors: 1 and itself. What numbers of different readouts are possible for a ring of prime length that uses at least two colors? Find a rule and explain it.

Problem 6: How many five-bead rings can be made when each bead may be any of three colors? Find a counting rule for c colors and any prime length p, and explain what it tells you about divisibility.